

HPS Gaussian-Process Resonance Search

Selected Reconstructed-MC Templates
Standalone Study v6.1

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Abstract

The 2021 Gaussian signal model is compared with normalized reconstructed target-constrained selected-candidate mass distributions from twelve A-prime MC masses between 40 and 260 MeV. The original $\pm 2.25\sigma_m$ fit and training mask, observed 10% spectrum, GP kernel states and coupling normalization are retained. Direct-template observed 90% CL_s limits and local p_0 values at the generated masses are compared with Gaussian fits to the same data. A dense morphing scan is reported separately as exploratory. The MC production is v13 and the observed sample is v16; their target constraint and momentum-sum labels agree, but timing selections differ and full production equivalence is not established. Stored reconstructed masses receive no added smearing or recentering. The inherited coupling coordinate is a conditional comparison, not a validated A-prime exclusion: an unmatched or accidental continuum need not scale with the signal coupling.

1 Observed comparison and scope

At the 10 shared generated masses, the median MC-to-Gaussian limit ratio is 0.9876; the range is 0.5332–3.7590. The median absolute relative limit change is 22.05%. The largest absolute relative limit change occurs at 60 MeV (+275.90%). The smallest direct-MC local probability among these tested points occurs at 80 MeV, with $p_0 = 2.477 \times 10^{-2}$ and $Z_0 = 1.964$. This is a sparse-grid local comparison, not a scan-global probability.

The direct comparison uses generated masses from 60 through 240 MeV in 20 MeV steps, inside the original 2021 search. The 40 and 260 MeV samples provide shape and interpolation information outside those direct test points. Full 2015 and full 2016 Gaussian references are retained in the package, but no matching reconstructed MC for those campaigns was supplied. The 2021 response is therefore not substituted into their fits.

The result is conditional on the supplied selected-MC shapes and their finite statistics. It is neither a newly qualified detector-response model nor a coverage calibration. The supplied candidate distributions have broad components and substantial displacement from the generated pole at several masses; they are not established as pure reconstructed resonance responses. An observed difference between the two curves can reflect mass bias, core width, non-Gaussian tails, selection differences and finite-MC fluctuations together.

2 Reconstructed mass distributions

The source histograms retain selected target-constrained masses within 0–400 MeV and record out-of-range counts separately. The pole mass sets the hypothesis; the reconstructed mean is allowed to differ. The comparison does not shift the MC peak onto the pole mass or force its width to the Gaussian parameterization.

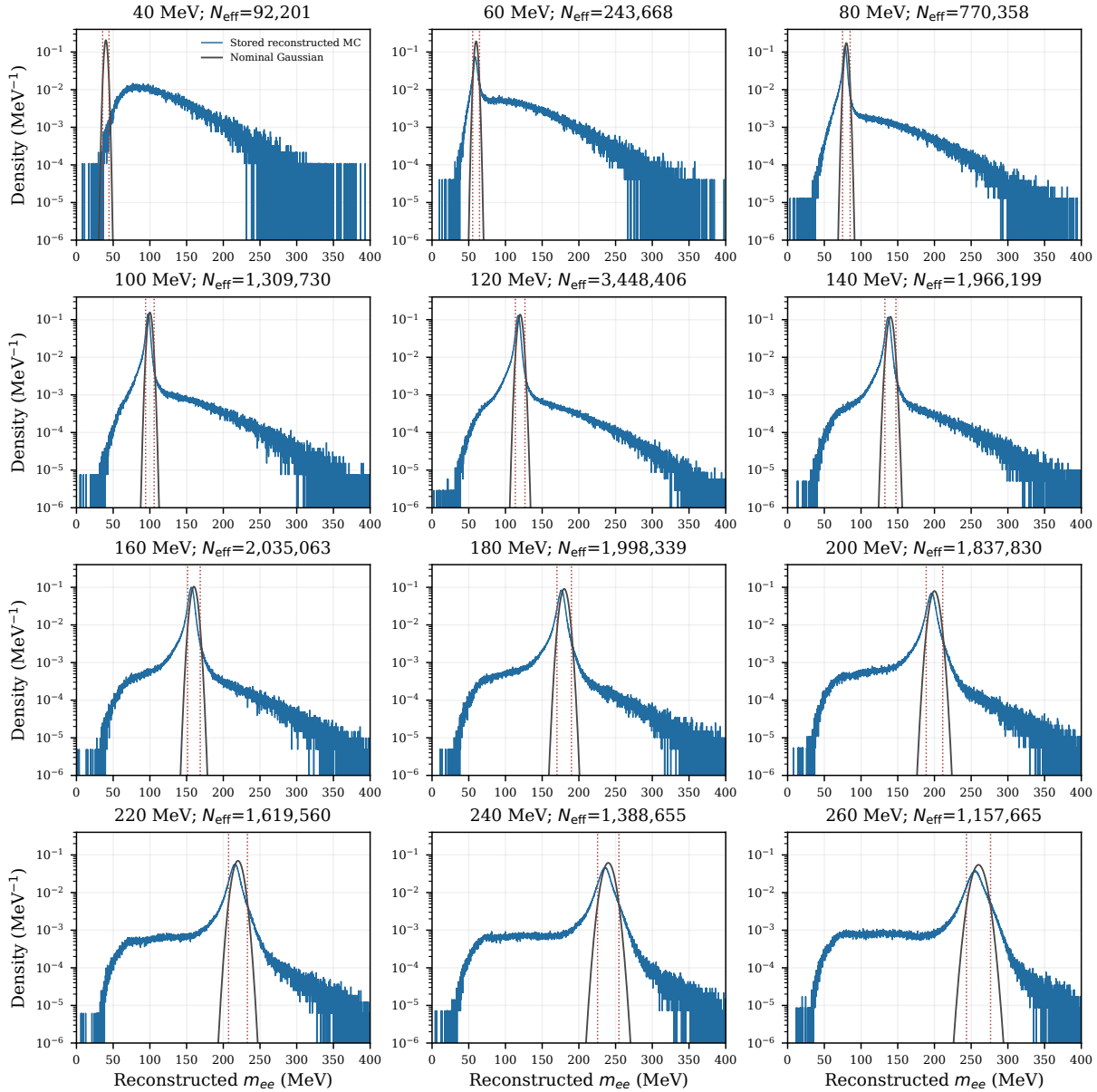


Figure 1: Unit-normalized stored reconstructed MC mass densities and the nominal Gaussian at each generated mass. Densities span the full stored range on a logarithmic axis so broad or displaced populations remain visible. Dotted lines mark the original $\pm 2.25\sigma_m$ boundaries, calculated from the nominal Gaussian resolution. Displayed densities are normalized over their stored histogram range; fitted templates retain their separately declared support normalization. No extra smearing or recentering is applied.

3 Selected MC statistics and shape properties

The source histograms store the weighted bin contents and squared-weight sums. The effective number of weighted entries is

$$N_{\text{eff}} = \frac{(\sum_j w_j)^2}{\sum_j w_j^2}. \quad (1)$$

It describes MC sampling, not data exposure or signal efficiency. Selected rows need not be unique physical events unless the production candidate policy establishes that identity. The source files, latest ROOT tree cycles, branches, weights and extraction counts are recorded in the histogram metadata and manifest.

The stored 0–400 MeV range contains 99.9848–99.9978% of the selected weight across samples; overflow counts are shown separately. All extracted weights are unit weights, so N_{eff} equals the selected count. The raw particle-vector momentum diagnostic falls below 2.8 GeV for 1.35–8.28% of rows. A bounded read finds that stored `psum` and `psum_scalar` satisfy the cut in every inspected row; the raw-vector discrepancy therefore reflects an incompatible convention, not an established selection failure.

m_0 (MeV)	Selected rows	Overflow	Median bias	σ_{68} (MeV)	Core (%)
40	92,201	2	+61.907	40.172	0.53
60	243,668	6	+7.500	40.204	39.18
80	770,358	18	-1.597	7.116	66.92
100	1,309,730	35	-2.411	3.724	72.61
120	3,448,406	123	-2.983	3.898	74.81
140	1,966,199	82	-3.510	4.441	75.64
160	2,035,063	109	-3.966	5.240	75.95
180	1,998,339	129	-4.382	6.311	75.86
200	1,837,830	141	-4.785	7.784	75.17
220	1,619,560	206	-5.250	9.868	73.84
240	1,388,655	180	-5.790	12.854	71.96
260	1,157,665	176	-6.581	17.255	69.90

Table 1: Selected reconstructed MC entries and saved shape summaries. Bias is the reconstructed median displacement, in MeV, from the generated pole. Median and quantile half-width σ_{68} are conditional on the stored 0–400 MeV range; the core probability uses the full selected weight. Overflow denotes selected rows above that range; it remains included in full-selected-yield bookkeeping.

The original 2021 resolution is held fixed when defining every mask:

$$\sigma_m(m) = 0.00184825 - 0.001375m + 0.085875m^2, \quad m, \sigma_m \text{ in GeV}. \quad (2)$$

Thus a narrower or shifted MC core changes the fraction of a unit signal inside the existing fitted bins. It does not silently redefine those bins or the sideband training set. The MC mass range and normalization support are recorded separately so losses at either boundary remain visible.

The word *smear*ed in the production directory is not sufficient to reconstruct the upstream smearing chain. The stored smear-ratio branches and cutflows are audited, while the template itself uses the stored target-constrained mass exactly once. No second factor of 1.25 or undocumented mass transformation is introduced.

4 Observed 2021 limits and local probabilities

The direct MC points use the same observed counts and correlated GP constraint as their Gaussian partners at each mass. The dense morph curve is displayed with a separate line style because its additional mass hypotheses require an interpolation assumption.

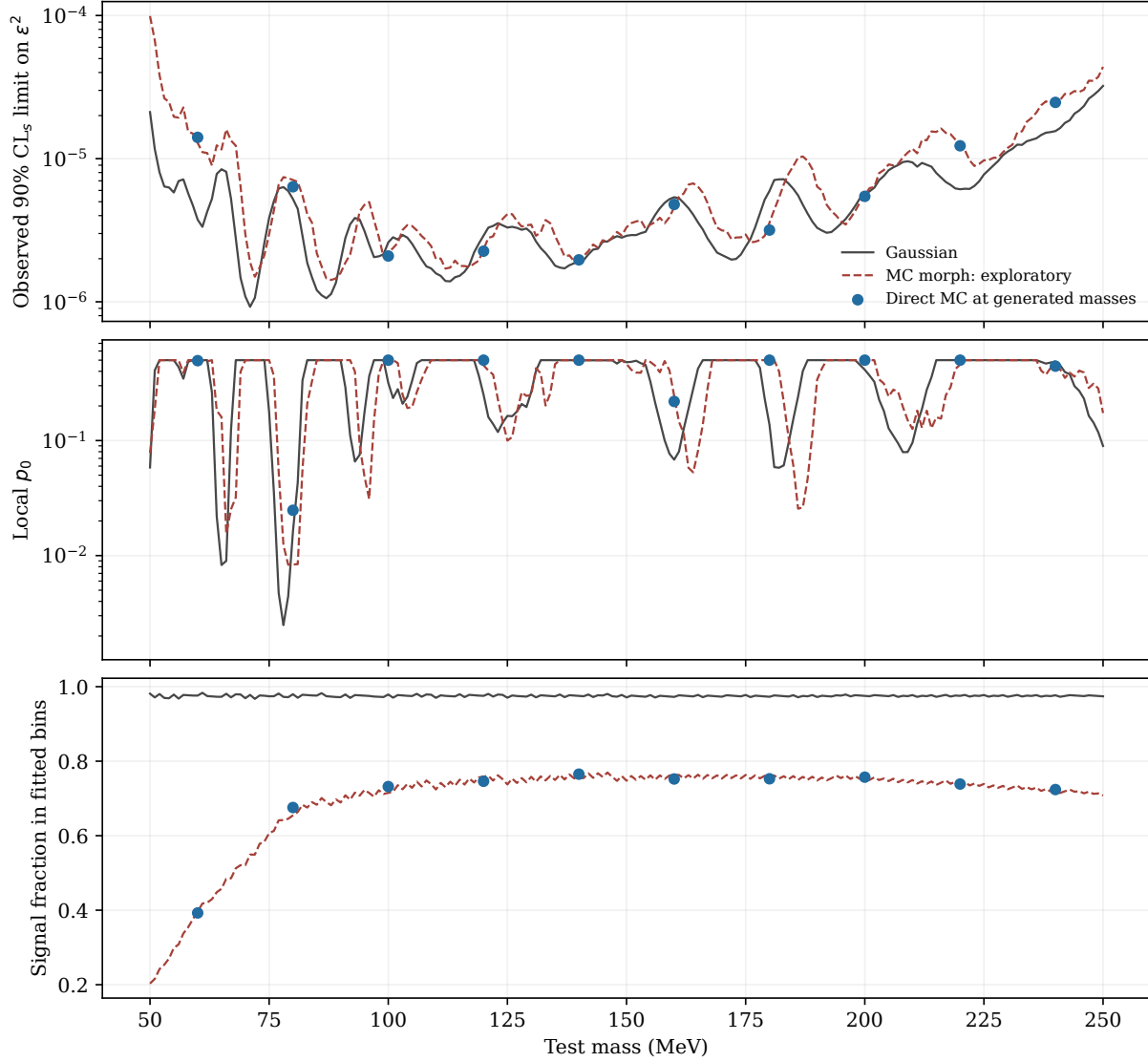


Figure 2: Observed 2021 10% limits, fixed-mass local probabilities and signal fractions in the original fitted bins. Points use the direct reconstructed templates at generated masses. The dashed curve, when present, uses exploratory MC morphing. The coupling limit includes the inherited minimal-visible branching correction once above the dimuon threshold.

5 Paired comparison at generated masses

The direct result is paired with the Gaussian at the same mass. The limit ratio is $R(m) = \varepsilon_{90,MC}^2/\varepsilon_{90,G}^2$, and the signed-root change is $\Delta r = r_{MC} - r_G$. Points at different masses share observations and GP training bins.

m_0	$\varepsilon_{90,G}^2$	$\varepsilon_{90,MC}^2$	Ratio	$p_{0,G}$	$p_{0,MC}$	Δr
60	3.747×10^{-6}	1.408×10^{-5}	3.7590	5.000×10^{-1}	4.950×10^{-1}	+0.825
80	5.221×10^{-6}	6.362×10^{-6}	1.2184	1.628×10^{-2}	2.477×10^{-2}	-0.174
100	2.599×10^{-6}	2.089×10^{-6}	0.8039	3.173×10^{-1}	5.000×10^{-1}	-0.969
120	2.905×10^{-6}	2.258×10^{-6}	0.7773	2.500×10^{-1}	5.000×10^{-1}	-0.969
140	1.968×10^{-6}	1.964×10^{-6}	0.9975	5.000×10^{-1}	5.000×10^{-1}	-0.356
160	5.376×10^{-6}	4.787×10^{-6}	0.8904	6.817×10^{-2}	2.188×10^{-1}	-0.713
180	5.946×10^{-6}	3.170×10^{-6}	0.5332	1.371×10^{-1}	5.000×10^{-1}	-2.194
200	5.577×10^{-6}	5.453×10^{-6}	0.9777	4.128×10^{-1}	5.000×10^{-1}	-0.684
220	6.114×10^{-6}	1.230×10^{-5}	2.0113	5.000×10^{-1}	5.000×10^{-1}	+1.117
240	1.558×10^{-5}	2.472×10^{-5}	1.5865	4.865×10^{-1}	4.436×10^{-1}	+0.108

Table 2: Direct-template comparison at generated masses. All probabilities are pointwise and use their stated signal template.

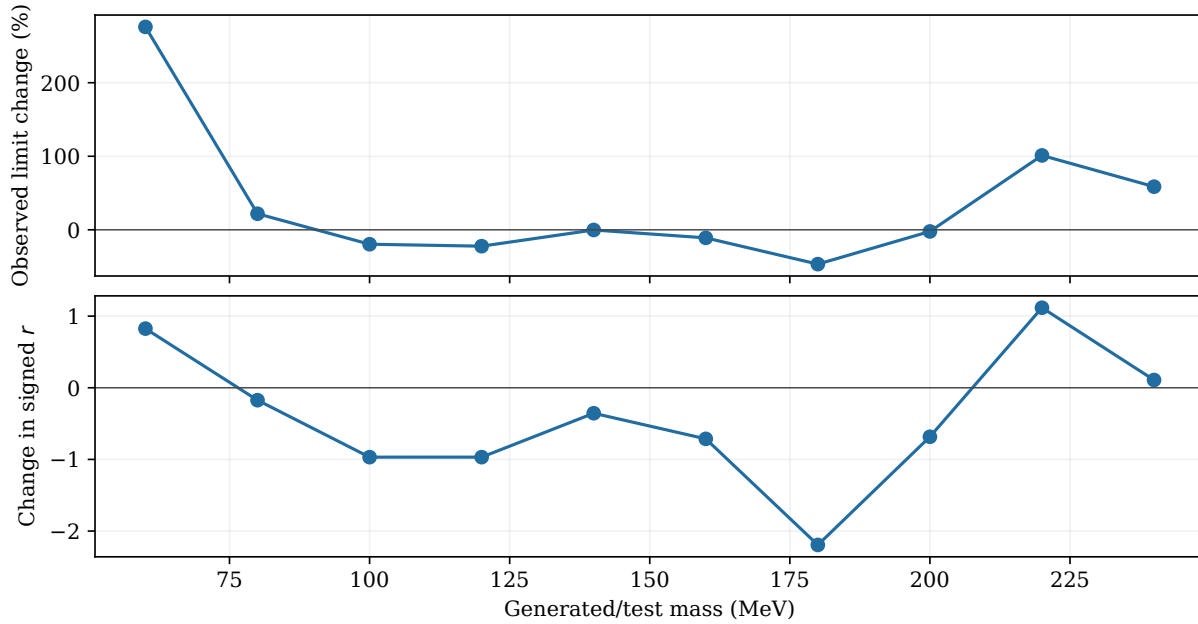


Figure 3: Observed fractional limit changes and signed-root differences for direct MC templates. These are paired descriptions of the same observed spectra. They are not independent residuals with a chi-squared sampling law.

6 Agreement summaries and interpolation boundary

Descriptive quantity	Direct MC versus Gaussian
Common generated masses	10
Median ratio	0.98760
16th–84th percentile ratio	0.78900–1.82440
Minimum / maximum ratio	0.53322 / 3.75896
Median absolute relative change	22.055%
RMS log ratio	0.54902
Within 1 / 5 / 10%	10.0% / 20.0% / 20.0%
Maximum $ \Delta r $	2.19398

Table 3: Descriptive agreement on the common direct-MC grid. Quantile ranges and percentages of agreeing masses are summaries of the curves, not confidence intervals or agreement probabilities.

A statistical probability for disagreement would require a specified joint sampling calculation that preserves the common data, overlapping sidebands and template estimation. An independent-point chi-squared test would omit those dependencies. The present comparison therefore reports sizes and locations of differences without assigning such a probability.

The reconstructed MC is available at discrete generated masses. A dense scan necessarily interpolates shape information. Holding out an interior generated mass and predicting its shape from its neighbors tests that interpolation against an available simulation sample. It does not test the underlying detector simulation against data, and it cannot establish sampling coverage of an upper limit.

The declared morph linearly mixes the neighboring CDFs in standardized residual $z = (m_{\text{rec}} - m_0)/\sigma_{\text{nominal}}(m_0)$, evaluating each input distribution at its own pole and nominal resolution. It uses no extrapolation beyond the supplied 40–260 MeV grid. Its dense observed scan covers the original 50–250 MeV search at 1 MeV spacing.

8 of 10 held-out shape checks pass the declared $D_{\text{CDF}} \leq 0.03$ and $|\Delta F_{\text{core}}| \leq 0.02$ criteria. The largest conditional-support CDF distance is 0.1395. The held-out limit ratios to direct MC range from 0.9142 to 1.3029. The dense curve remains exploratory regardless of this outcome.

The direct generated-mass results remain the primary comparison. Any minimum found on the denser exploratory grid depends on both the morphing prescription and the enlarged mass scan. Neither its local probability nor the smallest value selected across models is a global discovery probability.

6.1 Held-out shapes and finite-MC precision

m_0	D_{CDF}	ΔF_{core}	UL ratio	Δr	Shape check
60	0.1168	-0.0545	0.9142	+0.3724	fail
80	0.1395	-0.1090	1.3029	+0.0490	fail
100	0.0271	-0.0166	1.0540	+0.0364	pass
120	0.0118	-0.0062	1.0230	+0.0116	pass
140	0.0091	-0.0025	1.0117	+0.0037	pass
160	0.0057	-0.0016	1.0143	+0.0103	pass
180	0.0037	-0.0028	1.0165	+0.0124	pass
200	0.0033	-0.0030	1.0056	-0.0154	pass
220	0.0027	-0.0026	1.0177	+0.0289	pass
240	0.0052	-0.0006	1.0024	+0.0168	pass

Table 4: Leave-one-mass-out checks. CDF distances and core fractions use the conditional analysis-support distributions. Limit ratios and signed-root changes compare the held-out morph with direct MC at the same hypothesis. The shape thresholds are diagnostic criteria.

For each direct mass, 32 independent Poisson resamples of the stored unit-weight MC histogram bins are fitted against the fixed Gaussian reference. This conditional calculation quantifies template sampling precision under that bin-count model. It does not include detector systematics, selection differences, correlations between candidates or uncertainty in the observed-data background. No new observed-data experiments are generated.

m_0	Resamples	R_{16}	R_{50}	R_{84}	UL spread (%)
60	32	3.7284	3.7609	3.7925	1.063
80	32	1.2141	1.2173	1.2213	0.349
100	32	0.8016	0.8037	0.8061	0.302
120	32	0.7767	0.7772	0.7780	0.096
140	32	0.9960	0.9978	0.9987	0.125
160	32	0.8887	0.8908	0.8920	0.212
180	32	0.5320	0.5332	0.5350	0.243
200	32	0.9750	0.9773	0.9798	0.268
220	32	2.0067	2.0148	2.0185	0.312
240	32	1.5794	1.5852	1.5907	0.326

Table 5: Conditional finite-MC precision. R_{16}, R_{50}, R_{84} are quantiles of the MC-to-Gaussian upper-limit ratio across 32 MC-bin resamples. The last column is the relative standard deviation of the resampled limit around its central direct-template scale, in percent.

Thirty-two resamples resolve only a modest estimate of the central spread. The intervals are neither an observed-limit confidence band nor a significance for the discrepancy between templates.

7 Likelihood and normalization

The original primary window is used throughout. The GP trains only on bins outside $\pm 2.25\sigma_m$, using archived kernel coordinates. Within a mass hypothesis, both signal models share its predicted count mean b , covariance C and observed counts n . With $LL^T = C$,

$$a = \frac{\varepsilon^2}{10^{-8}}, \quad T_i = K(m) 10^{-8} w_i, \quad (3)$$

$$\lambda_i = b_i + (L\theta)_i + aT_i, \quad (4)$$

$$\ell(a, \theta) = \sum_i \left[\lambda_i - n_i + n_i \log \frac{n_i}{\lambda_i} \right] + \frac{1}{2} \theta^T \theta. \quad (5)$$

The zero-count contribution is λ_i . The correlated background deformation is profiled at every tested amplitude. The bin probabilities w_i are normalized over the declared full support before selecting fitted bins; their restriction to the primary window is not renormalized.

For MC the full fitting support is 36–299.75 MeV. Its retained fraction f_{support} is recorded separately. The ledger’s `full_yield_90` is the support-normalized yield; the additional `full_selected_yield_90` divides it by f_{support} . That bookkeeping conversion does not establish a calibrated coupling normalization outside the analysis support.

Here $K(m)$ is the unchanged prompt-density conversion. The solver’s `A90` column is a_{90} , while the support-normalized event yield is $N_{90} = a_{90} K(m) 10^{-8} = K(m) \varepsilon_{90}^2$. Above $2m_\mu$, the inherited minimal-visible conversion multiplies the electron-only coupling limit by $1 + \sqrt{1 - 4r_\mu(1 + 2r_\mu)}$, with $r_\mu = (m_\mu/m)^2$. This correction does not change the visible yield or local probability. Density continues to use the nominal $m \pm 1.64\sigma_m$ native-bin interval.

The observed limit solves the bounded, piecewise-asymptotic $CL_s(a_{90}) = 0.10$, using explicitly profiled observed and background-Asimov likelihood ratios [2]. The free diagnostic amplitude may be negative while physical Poisson expectations are maintained. Writing its signed likelihood root as r ,

$$Z_0 = \max(r, 0), \quad p_0 = \Phi(-Z_0). \quad (6)$$

Deficits therefore have the inherited reference value $p_0 = 0.5$. This is a local asymptotic reference, not the probability of the most significant result anywhere in a scan.

The Gaussian and MC comparisons use the same converged solver. The historical v5.9 audit found a stopping discrepancy in an older optimizer; the present paired comparison does not mix that older table with newly computed MC fits. Numerical convergence and finite-MC statistical qualification are separate questions.

8 Selection provenance and interpretation

The supplied MC directory is `/sdf/data/hps/physics2021/preselection/v13/ap_signal_prompt_smeared`. The observed input derives from `/sdf/data/hps/physics2021/preselection/v16/data_10pc_prompt_TC_psum2p8`. The MC uses the stored `vertex./vertex.invM_` branch and the target-constrained vertex category. Production labels agree on the target constraint and $p_{\text{sum}} > 2.8$ GeV, but the inspected MC timing windows include 5.7 and 9.2 ns where the observed sample uses 5.1 and 7.8 ns. Full production equivalence is not established. No new selection is silently imposed to make the two samples appear matched.

A bounded inspection finds both stored daughter truth-link flags false for every inspected row. Those flags cannot identify matched signal daughters, and their values alone do not prove that the reconstructed candidates lack signal daughters. No additional truth selection is defensible from them. The fit therefore substitutes the supplied all-selected candidate distribution conditionally.

The stored track hit counters are three-dimensional; cutflow labels refer to two-dimensional-hit requirements. Those labels are not a license to apply their thresholds directly to the stored hit counters. The local and remote selection audits preserve the evidence and the unresolved production details. ROOT files can contain older tree cycles; only the latest `preselection` cycle contributes to extraction.

Input files and configurations are identified by saved paths, sizes and SHA-256 hashes. Extraction uses bounded chunks and retains bin sums of weights and squared weights. All results, scripts, reconstructed-mass histograms and report sources are mirrored to `/sdf/home/e/epeets/src/hps_gpr_v6p1_mc_signal_templates_20260922`. The original observed result and source ROOT files are preserved.

This study measures how the selected reconstructed 2021 MC changes the conditional observed inference under the original background model and mask. It does not isolate a pure tail uncertainty or establish that either signal template is the correct detector response. Selection differences, finite MC and interpolation remain explicit qualifications. No MC-template claim is made for the 2015 or 2016 detector configurations.

Approximately 24–61% of the direct-mass selected signal distributions lie outside the primary blind window. Signal in those training bins can be absorbed by a background-only GP. This observed template replacement keeps the sideband prediction fixed; it does not model signal-contamination feedback, extraction recovery or coverage for these broad distributions.

References

- [1] E. Peets, M. Gignac and E. Hsu, *HPS Gaussian-Process Resonance Search: Analysis Note v5.0.5*, 16 September 2026, internal review draft and archived source package.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, “Asymptotic formulae for likelihood-based tests of new physics,” *Eur. Phys. J. C* **71**, 1554 (2011), arXiv:1007.1727.